Leo Zhao

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Education

University of Waterloo *September 2025 – April 2030*
Candidate for Bachelor of Applied Science in Mechatronics Engineering
GPA: 4.0/4.0
Relevant Coursework: CAD Fundamentals, C++ Fundamentals, Introduction to Mechatronics, Circuits, Materials
Extracurriculars: Academic Representative for 120 students, Waterloo Aerial Robotics Group Mechanical Member
Awards: Dean's Scholar (Highest Academic Average in my Program) in 1B

Technical Skills

Programming Languages: Java, Python, C++, HTML, CSS
Developer Tools: AutoCAD, SolidWorks, Onshape, GD&T, GitHub, Arduino, VS Code, Agile, Spyder, IntelliJ, Eclipse
Office Tools: Excel, Word, Powerpoint, LaTex
Certifications: CSWA SolidWorks Asscoiate, LinkedIn's Excel Essential Training
Languages: English (native), French (B2), Chinese (proficient)

Projects

Openbooks Accounting System | *Python, SQLite, HTML, CSS* February 2026
- Built full double-entry accounting app with SQLite, packaged as a PyInstaller Windows .exe
- Implemented chart of accounts, journal entries, general ledger, balance sheet, income statement, and PDF transaction scanner using python
- Added tax rates, date-range reporting, and real-time dashboard updates ensuring reports always reflect current balances

Self Sorting Pool Table (Bttrpie) | *SolidWorks, C++, GitHub* October 2025 – December 2025
- Developed a miniature pool table featuring an automated ball sorting system powered by a 3-axis gantry and a grabbing claw
- Modeled and simulated all mechanical assemblies in SolidWorks, including motion testing
- Programmed supplementary C++ control software enabling manual operation of multi-axis movement, claw control, and positional feedback
- Machined and fabricated components for the structural frame and mechanical systems

Noise Cancelling Speaker | *Arduino, C++* September 2025
- Collaborated in a team to design and build a functional speaker capable of producing sound levels up to 80 dB using Arduino
- Optimized enclosure geometry to direct and focus audio output for improved sound projection
- Programmed C++ control logic enabling multiple tune playback via user-input buttons

Experience

Manufacturing Engineering Intern at Linamar *May 2026 – Present*
Guelph, ON
- Produced over 130 AutoCAD drawings and 100 SolidWorks deliverables, translating manufacturing needs into accurate models, assemblies, and technical documentation with proper GD&T tolerances
- Designed an in-table pneumatic gauge in SolidWorks to verify spline geometry and depth, automate part ejection, and reduce operator strain
- Investigated a recurring burr-quality issue through an analysis of data and authored a report contributing to documentation for an SR&ED claim exceeding $2 000

Waterloo Aerial Robotics Group (WARG) Mechanical Member *December 2025 - Present*
Waterloo, ON
- Redesigned a SolidWorks gimbal camera mount by removing non-critical material, reducing mass by approximately 33% while maintaining functional requirements for aerial imaging.
- Used XFLR5 to model Helios, a fixed-wing aircraft, and extract aileron, elevator, and rudder control derivatives to guide control-surface sizing and aerodynamic design decisions.
- Co-designed in SolidWorks and fabricated a reliable payload-release mechanism that consistently deployed the team's largest competition payload, a mock ladder, during flight operations.